



Lecture 8: Distribution Systems

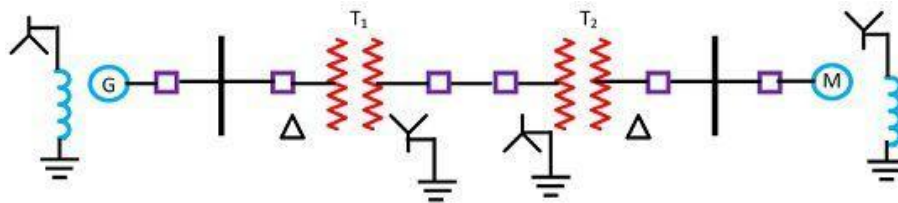
EEE 321: Power System I

tareq.aziz@bracu.ac.bd

Presentation Outline

- ❑ Introduction to Distribution Systems
- ❑ Classification of Distribution Systems
- ❑ AC Distribution System
- ❑ Connection schemes of Distribution System
- ❑ DC Distribution Calculations
- ❑ AC Distribution Calculations
- ❑ Examples

Introduction to Distribution System



Single Line Representation of a Typical Power System

Circuit Globe

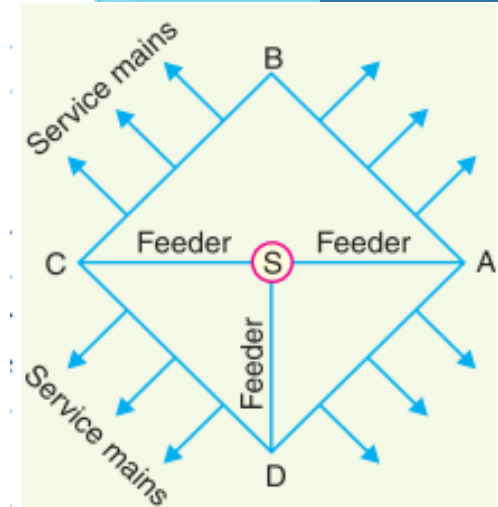


Fig. 12.1

That part of power system that distributes electric power for local use is known as distribution system.

In general, the distribution system is the electrical system between the sub-station fed by the transmission system and the consumers meters. It generally consists of *feeders*, *distributors* and the *service mains*. Fig. 12.1 shows the single line diagram of a typical low tension distribution system.

(i) *Feeders*. A feeder is a conductor which connects the sub-station (or localised generating station) to the area where power is to be distributed. Generally, no tappings are taken from the feeder so that current in it remains the same throughout. The main consideration in the design of a feeder is the current carrying capacity.

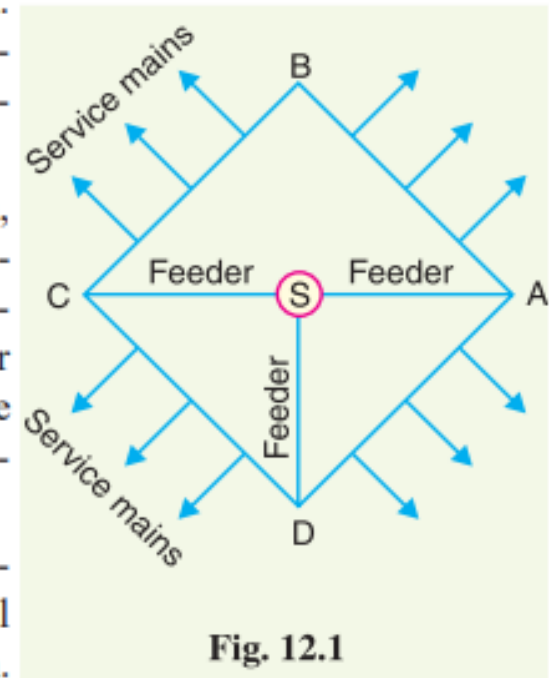
(ii) *Distributor*. A distributor is a conductor from which tappings are taken for supply to the consumers. In Fig. 12.1, *AB*, *BC*, *CD* and *DA* are the distributors. The current through a distributor is not constant because tappings are taken at various places along its length. While designing a distributor, voltage drop along its length is the main consideration since the statutory limit of voltage variations is $\pm 6\%$ of rated value at the consumers' terminals.

(iii) *Service mains*. A service mains is generally a small cable which connects the distributor to the consumers' terminals.

Classification of Distribution Systems

A distribution system may be classified according to ;

- (i) **Nature of current.** According to nature of current, distribution system may be classified as (a) d.c. distribution system (b) a.c. distribution system. Now-a-days, a.c. system is universally adopted for distribution of electric power as it is simpler and more economical than direct current method.
- (ii) **Type of construction.** According to type of construction, distribution system may be classified as (a) overhead system (b) underground system. The overhead system is generally employed for distribution as it is 5 to 10 times cheaper than the equivalent underground system. In general, the underground system is used at places where overhead construction is impracticable or prohibited by the local laws.
- (iii) **Scheme of connection.** According to scheme of connection, the distribution system may be classified as (a) radial system (b) ring main system (c) inter-connected system. Each scheme has its own advantages and disadvantages and those are discussed in Art.12.7.

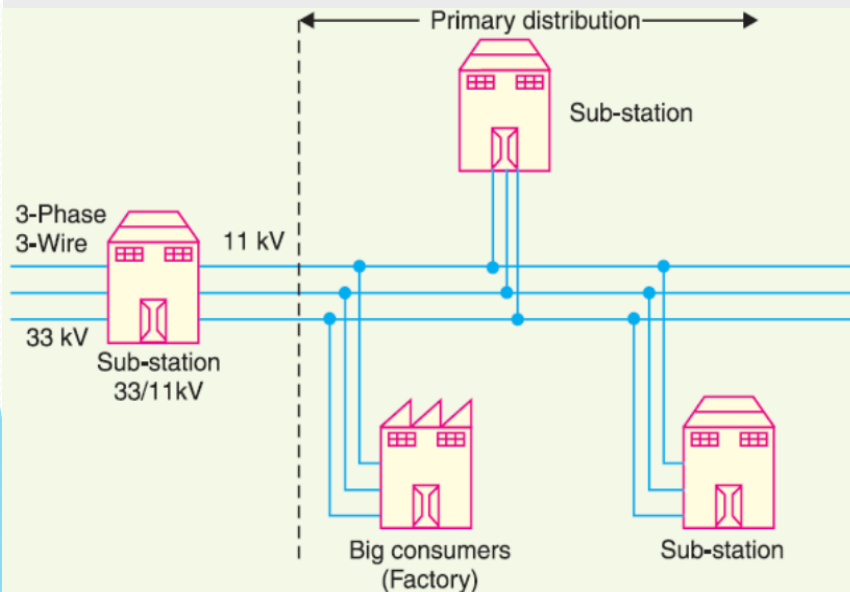


AC Distribution Systems

AC power can be distributed at a high voltage through a transformer. High transmission and distribution voltages have greatly reduced the current in the conductors and the resulting line losses. The ac distribution system can be classified into:

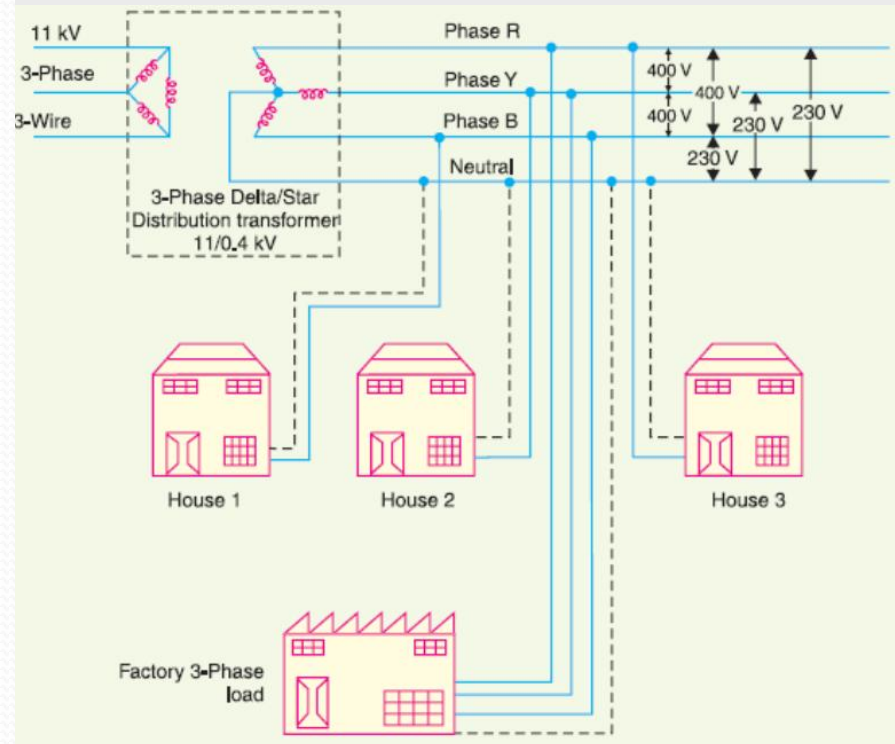
Primary Distribution System

This part of distribution system generally operates at somewhat higher voltages and handle large blocks of electrical energy. The most commonly used primary distribution voltages are **11 KV, 6.6 KV, and 3.3 KV**. Due to economic considerations, primary distribution system is carried out by 3 phase 3 wire system.



Secondary Distribution System

The most commonly used secondary distribution voltages are **400/230 volts**. Secondary distribution system is carried out by 3 phase 4 wire system.



Connection schemes of Distribution System

(i) **Radial System.** In this system, separate feeders radiate from a single substation and feed the distributors at one end only. Fig. 12.8 (i) shows a single line diagram of a radial system for d.c. distribution where a feeder OC supplies a distributor AB at point A . Obviously, the distributor is fed at one end only *i.e.*, point A is this case. Fig. 12.8 (ii) shows a single line diagram of radial system for a.c. distribution. The radial system is employed only when power is generated at low voltage and the substation is located at the centre of the load.

This is the simplest distribution circuit and has the lowest initial cost. However, it suffers from the following drawbacks :

- (a) The end of the distributor nearest to the feeding point will be heavily loaded.
- (b) The consumers are dependent on a single feeder and single distributor. Therefore, any fault on the feeder or distributor cuts off supply to the consumers who are on the side of the fault away from the substation.
- (c) The consumers at the distant end of the distributor would be subjected to serious voltage fluctuations when the load on the distributor changes.

Due to these limitations, this system is used for short distances only.

(ii) **Ring main system.** In this system, the primaries of distribution transformers form a loop. The loop circuit starts from the substation bus-bars, makes a loop through the area to be served, and returns to the substation. Fig. 12.9 shows the single line diagram of ring main system for a.c. distribution where substation supplies to the closed feeder $LMNOPQRS$. The distributors are tapped from different points M , O and Q of the feeder through distribution transformers. The ring main system has the following advantages :

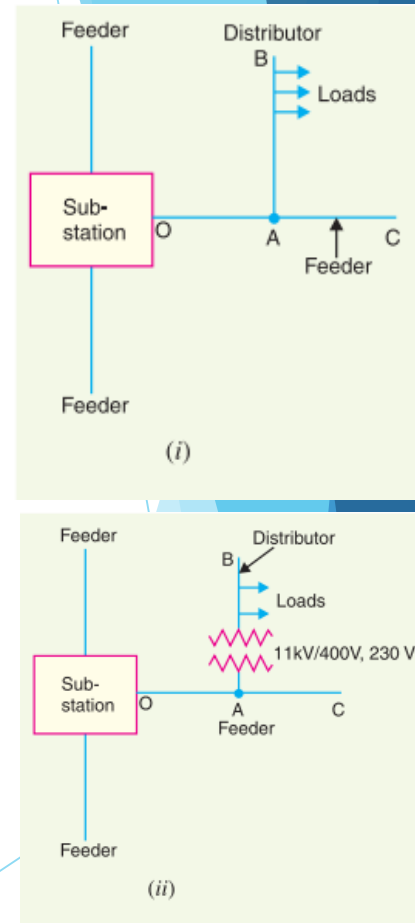
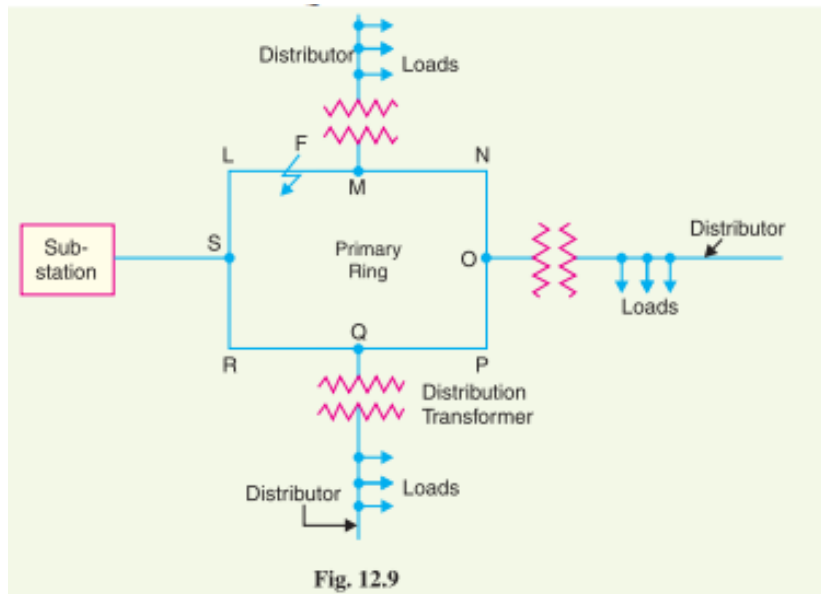


Fig. 12.8

Connection schemes of Distribution System

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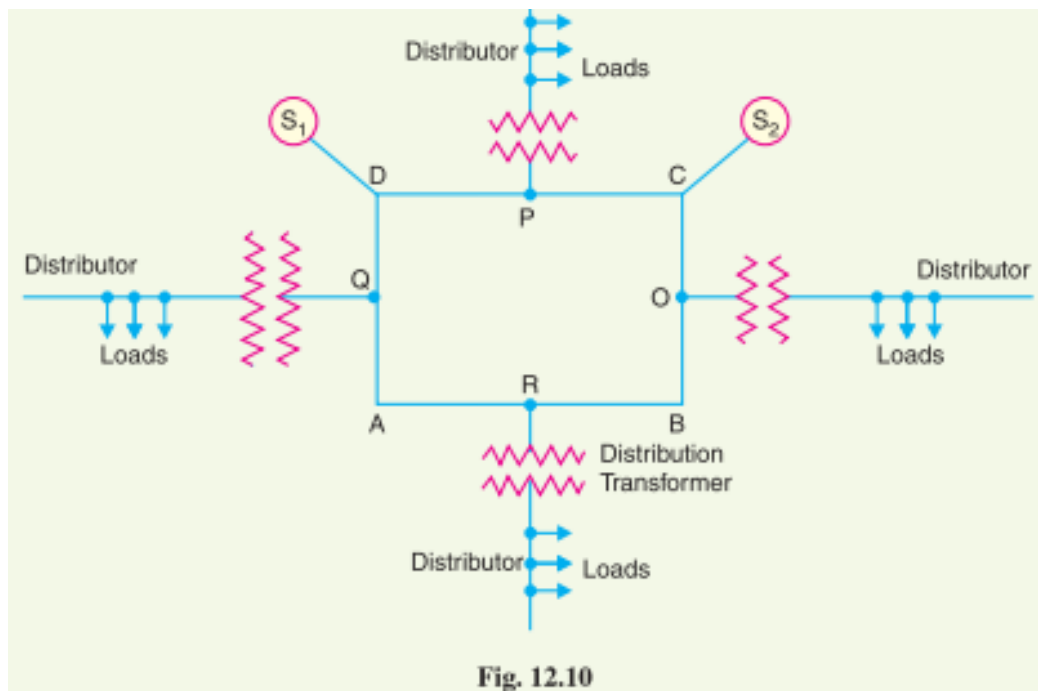
- (a) There are less voltage fluctuations at consumer's terminals.
- (b) The system is very reliable as each distributor is fed *via* *two feeders. In the event of fault on any section of the feeder, the continuity of supply is maintained. For example, suppose that fault occurs at any point F of section SLM of the feeder. Then section SLM of the feeder can be isolated for repairs and at the same time continuity of supply is maintained to all the consumers *via* the feeder $SRQPONM$.



Connection schemes of Distribution System

(iii) **Interconnected system.** When the feeder ring is energised by two or more than two generating stations or substations, it is called inter-connected system. Fig. 12.10 shows the single line diagram of interconnected system where the closed feeder ring $ABCD$ is supplied by two substations S_1 and S_2 at points D and C respectively. Distributors are connected to points O , P , Q and R of the feeder ring through distribution transformers. The interconnected system has the following advantages :

- (a) It increases the service reliability.
- (b) Any area fed from one generating station during peak load hours can be fed from the other generating station. This reduces reserve power capacity and increases efficiency of the system.



DC Distribution Calculations

Fig. 13.5 shows the single line diagram of a 2-wire d.c. distributor AB fed at one end A and having concentrated loads I_1, I_2, I_3 and I_4 tapped off at points C, D, E and F respectively.

Let r_1, r_2, r_3 and r_4 be the resistances of both wires (go and return) of the sections AC, CD, DE and EF of the distributor respectively.

$$\text{Current fed from point } A = I_1 + I_2 + I_3 + I_4$$

$$\text{Current in section } AC = I_1 + I_2 + I_3 + I_4$$

$$\text{Current in section } CD = I_2 + I_3 + I_4$$

$$\text{Current in section } DE = I_3 + I_4$$

$$\text{Current in section } EF = I_4$$

$$\text{Voltage drop in section } AC = r_1 (I_1 + I_2 + I_3 + I_4)$$

$$\text{Voltage drop in section } CD = r_2 (I_2 + I_3 + I_4)$$

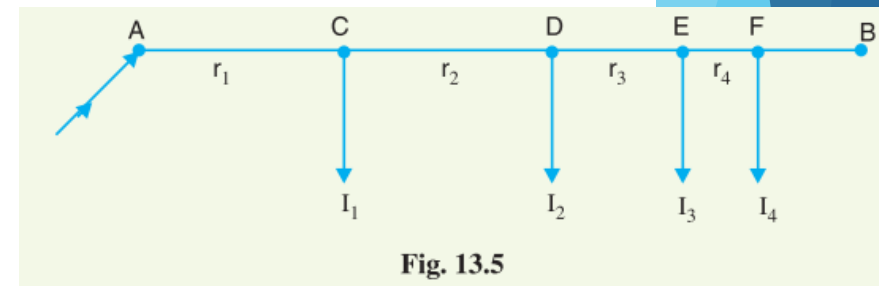
$$\text{Voltage drop in section } DE = r_3 (I_3 + I_4)$$

$$\text{Voltage drop in section } EF = r_4 I_4$$

$\therefore$ Total voltage drop in the distributor

$$= r_1 (I_1 + I_2 + I_3 + I_4) + r_2 (I_2 + I_3 + I_4) + r_3 (I_3 + I_4) + r_4 I_4$$

It is easy to see that the minimum potential will occur at point F which is farthest from the feeding point A .



AC Distribution Calculations

A.C. distribution calculations differ from those of d.c. distribution in the following respects :

- (i) In case of d.c. system, the voltage drop is due to resistance alone. However, in a.c. system, the voltage drops are due to the combined effects of resistance, inductance and capacitance.
- (ii) In a d.c. system, additions and subtractions of currents or voltages are done arithmetically but in case of a.c. system, these operations are done vectorially.
- (iii) In an a.c. system, power factor (p.f.) has to be taken into account. Loads tapped off from the distributor are generally at different power factors. There are two ways of referring power factor *viz*
 - (a) It may be referred to supply or receiving end voltage which is regarded as the reference vector.
 - (b) It may be referred to the voltage at the load point itself.

There are several ways of solving a.c. distribution problems. However, symbolic notation method has been found to be most convenient for this purpose. In this method, voltages, currents and impedances are expressed in complex notation and the calculations are made exactly as in d.c. distribution.

AC Distribution Calculations

(i) **Power factors referred to receiving end voltage.** Consider an a.c. distributor AB with concentrated loads of I_1 and I_2 tapped off at points C and B as shown in Fig. 14.1. Taking the receiving end voltage V_B as the reference vector, let lagging power factors at C and B be $\cos \phi_1$ and $\cos \phi_2$ w.r.t. V_B . Let R_1, X_1 and R_2, X_2 be the resistance and reactance of sections AC and CB of the distributor.

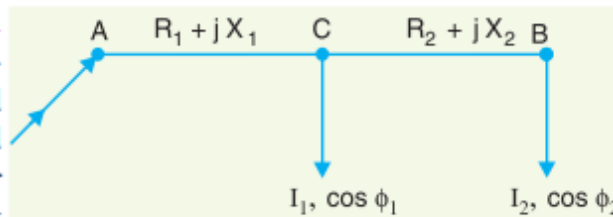


Fig. 14.1

Impedance of section AC , $\overrightarrow{Z_{AC}} = R_1 + j X_1$

Impedance of section CB , $\overrightarrow{Z_{CB}} = R_2 + j X_2$

Load current at point C , $\overrightarrow{I_1} = I_1 (\cos \phi_1 - j \sin \phi_1)$

Load current at point B , $\overrightarrow{I_2} = I_2 (\cos \phi_2 - j \sin \phi_2)$

Current in section CB , $\overrightarrow{I_{CB}} = \overrightarrow{I_2} = I_2 (\cos \phi_2 - j \sin \phi_2)$

Current in section AC , $\overrightarrow{I_{AC}} = \overrightarrow{I_1} + \overrightarrow{I_2}$
 $= I_1 (\cos \phi_1 - j \sin \phi_1) + I_2 (\cos \phi_2 - j \sin \phi_2)$

Voltage drop in section CB , $\overrightarrow{V_{CB}} = \overrightarrow{I_{CB}} \overrightarrow{Z_{CB}} = I_2 (\cos \phi_2 - j \sin \phi_2) (R_2 + j X_2)$

Voltage drop in section AC , $\overrightarrow{V_{AC}} = \overrightarrow{I_{AC}} \overrightarrow{Z_{AC}} = (\overrightarrow{I_1} + \overrightarrow{I_2}) \overrightarrow{Z_{AC}}$
 $= [I_1 (\cos \phi_1 - j \sin \phi_1) + I_2 (\cos \phi_2 - j \sin \phi_2)] [R_1 + j X_1]$

Sending end voltage, $\overrightarrow{V_A} = \overrightarrow{V_B} + \overrightarrow{V_{CB}} + \overrightarrow{V_{AC}}$

Sending end current, $\overrightarrow{I_A} = \overrightarrow{I_1} + \overrightarrow{I_2}$

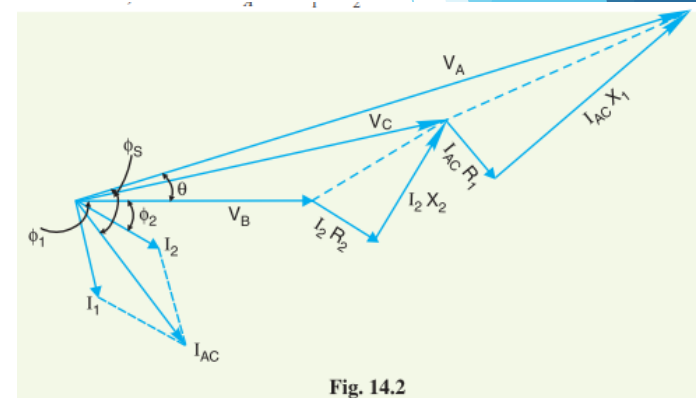


Fig. 14.2

The vector diagram of the a.c. distributor under these conditions is shown in Fig. 14.2. Here, the receiving end voltage V_B is taken as the reference vector. As power factors of loads are given w.r.t. V_B , therefore, I_1 and I_2 lag behind V_B by ϕ_1 and ϕ_2 respectively.

Examples

Example 14.1. A single phase a.c. distributor AB 300 metres long is fed from end A and is loaded as under :

(i) 100 A at 0.707 p.f. lagging 200 m from point A

(ii) 200 A at 0.8 p.f. lagging 300 m from point A

The load resistance and reactance of the distributor is 0.2Ω and 0.1Ω per kilometre. Calculate the total voltage drop in the distributor. The load power factors refer to the voltage at the far end.

Solution. Fig. 14.4 shows the single line diagram of the distributor.

Impedance of distributor/km $= (0.2 + j 0.1) \Omega$

Impedance of section AC, $\overline{Z}_{AC} = (0.2 + j 0.1) \times 200/1000 = (0.04 + j 0.02) \Omega$

Impedance of section CB, $\overline{Z}_{CB} = (0.2 + j 0.1) \times 100/1000 = (0.02 + j 0.01) \Omega$

Taking voltage at the far end B as the reference vector, we have,

Load current at point B, $\vec{I}_2 = I_2 (\cos \phi_2 - j \sin \phi_2) = 200 (0.8 - j 0.6)$
 $= (160 - j 120) \text{ A}$

Load current at point C, $\vec{I}_1 = I_1 (\cos \phi_1 - j \sin \phi_1) = 100 (0.707 - j 0.707)$
 $= (70.7 - j 70.7) \text{ A}$

Current in section CB, $\overline{I}_{CB} = \vec{I}_2 = (160 - j 120) \text{ A}$

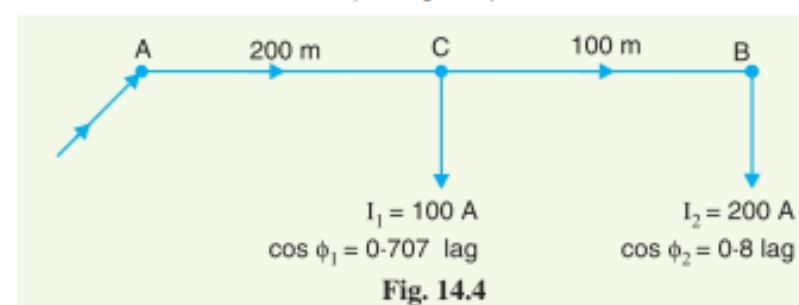
Current in section AC, $\overline{I}_{AC} = \vec{I}_1 + \vec{I}_2 = (70.7 - j 70.7) + (160 - j 120)$
 $= (230.7 - j 190.7) \text{ A}$

Voltage drop in section CB, $\overline{V}_{CB} = \overline{I}_{CB} \overline{Z}_{CB} = (160 - j 120) (0.02 + j 0.01)$
 $= (4.4 - j 0.8) \text{ volts}$

Voltage drop in section AC, $\overline{V}_{AC} = \overline{I}_{AC} \overline{Z}_{AC} = (230.7 - j 190.7) (0.04 + j 0.02)$
 $= (13.04 - j 3.01) \text{ volts}$

Voltage drop in the distributor $= \overline{V}_{AC} + \overline{V}_{CB} = (13.04 - j 3.01) + (4.4 - j 0.8)$
 $= (17.44 - j 3.81) \text{ volts}$

Magnitude of drop $= \sqrt{(17.44)^2 + (3.81)^2} = 17.85 \text{ V}$

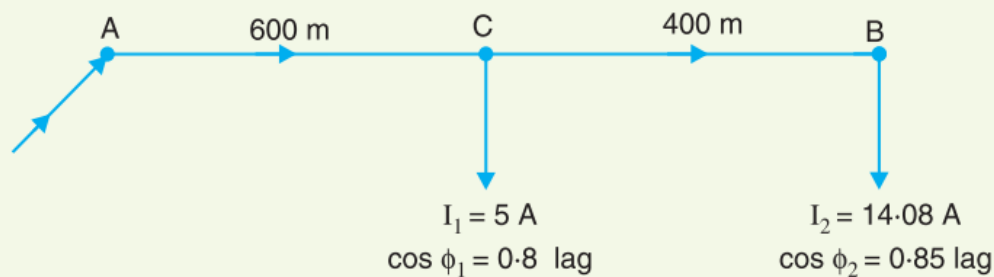


Please check
Example 14.2, 14.3,
Mehta

Practice Problems

Example 14.5. A 3-phase, 400V distributor AB is loaded as shown in Fig.14.8. The 3-phase load at point C takes 5A per phase at a p.f. of 0.8 lagging. At point B, a 3-phase, 400 V induction motor is connected which has an output of 10 H.P. with an efficiency of 90% and p.f. 0.85 lagging.

If voltage at point B is to be maintained at 400 V, what should be the voltage at point A ? The resistance and reactance of the line are 1Ω and 0.5Ω per phase per kilometre respectively.



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Example 14.2,14.3,
Mehta

Answer: 433V

Reference:

'Principles of Power System', V.K. Mehta and Rohit Mehta



QUESTIONS AND DISCUSSIONS

For further queries: tareq.aziz@bracu.ac.bd